Supplementary Material for "Excitation-Dependent Identifiability of Latent Higher-Order Structure from Edge Flows"

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This supplement extends the main paper in five directions: **S1** Fano-type converses for *joint* support recovery (including a signal-distribution-agnostic bound and a sub-Gaussian achievability counterpart); **S2** identifiability under *partial edge observation*; **S3** *plug-in* estimation of the variance parameters and a fully adaptive detector; **S4** full proofs of the main paper's lifted-spark lemma, the three excitation regimes, separation/exponent theorem, and NNLS consistency/failure bound, with exhaustive numerical verification; **S5** construction details of the cyclone vortex-localization study. The released test-suite provides numerical guardrails (not proofs) for the quoted constants; Section S6 maps every figure to its script. Notation follows the main paper: N snapshots, candidate set $\mathcal{T}$ with $p = |\mathcal{T}|$, latent support $\mathcal{S}$ with $|\mathcal{S}| = k$, curl-SNR $\rho = 3\sigma_c^2/\sigma_n^2$, per-coordinate null/alternative variances $v_0 = 3\sigma_n^2$, $v_1 = v_0(1+\rho)$. Throughout S1–S2 the candidates are pairwise edge-disjoint, so the curl coordinates are independent given $\mathcal{S}$ and the triangle Gram matrix is $\boldsymbol{G} = 3\boldsymbol{I}$.

S1 Fano-type converses for joint support recovery

The main paper's Prop. 2 contains a *single-triangle* converse — the floor $\rho^\star(N) = \Theta(\sqrt{\log(1/\delta)/N})$ — together with the Gaussian Fano bound for *joint* recovery of the whole support; Theorem S1.2 below proves the latter, and the signal-agnostic variant extends it.

We first record a reduction used by both proofs.

Lemma S1.1 (Sufficiency of the curl coordinates)**.** *Consider the model of the main paper (Eq. (1)) with isotropic Gaussian noise $\boldsymbol{n}_t \sim \mathcal{N}(0, \sigma_n^2\boldsymbol{I})$, mutually independent $(\boldsymbol{a}_t, \boldsymbol{y}_t, \boldsymbol{h}_t, \boldsymbol{n}_t)$, and* arbitrary *laws for $(\boldsymbol{a}_t, \boldsymbol{y}_t, \boldsymbol{h}_t)$ (Gaussianity is needed only for the noise). Then*

$$I(\mathcal{S}; \{\boldsymbol{f}_t\}_{t=1}^N) \;=\; I(\mathcal{S}; \{\boldsymbol{c}_t\}_{t=1}^N), \qquad \boldsymbol{c}_t = \boldsymbol{B}_2^\top \boldsymbol{f}_t.$$

Proof. Decompose $\boldsymbol{f}_t$ orthogonally into its gradient, curl, and harmonic components. Because the isotropic Gaussian noise has independent projections onto orthogonal subspaces, and the gradient (resp. harmonic, curl) component of $\boldsymbol{f}_t$ equals $\boldsymbol{B}_1^\top \boldsymbol{a}_t$ (resp. $\boldsymbol{h}_t$, $\boldsymbol{B}_{2,\mathcal{S}}\boldsymbol{y}_t$) plus the corresponding independent noise projection, the three components are mutually independent, and only the curl component depends on $\mathcal{S}$. Hence the gradient and harmonic components are independent of $(\mathcal{S}, \text{curl component})$ and carry zero information about $\mathcal{S}$. Finally, on edge-disjoint candidates $\boldsymbol{B}_2$ has full column rank, so $\boldsymbol{c}_t$ is a bijective linear image of the curl component. □

Theorem S1.2 (Gaussian Fano converse)**.** *Let $\mathcal{S}$ be uniform on the $\binom{p}{k}$ supports of size k, with Gaussian signals and noise and edge-disjoint candidates. If an estimator achieves* $\Pr[\hat{\mathcal{S}} \neq \mathcal{S}] \leq \varepsilon$, *then*

$$N \;\geq\; \frac{(1-\varepsilon)\log\binom{p}{k} - \log 2}{k\,\mathrm{KL}_1(\rho)}, \qquad \mathrm{KL}_1(\rho) = \tfrac{1}{2}\big(\rho - \log(1+\rho)\big) \;\leq\; \tfrac{\rho^2}{4}. \tag{1}$$

In particular $\rho^2 N \gtrsim 4(1-\varepsilon)\log(p/k)$*: joint recovery pays a* $\log\binom{p}{k}$ *factor where the single-triangle test pays* $\log(1/\delta)$*.*

Proof. By Lemma S1.1 and Fano's inequality, $\Pr[\hat{\mathcal{S}} \neq \mathcal{S}] \geq 1 - (I(\mathcal{S}; \boldsymbol{C}^N) + \log 2)/\log\binom{p}{k}$ with $\boldsymbol{C}^N = \{\boldsymbol{c}_t\}$. Bound the mutual information against the all-inactive reference law P_0: $I(\mathcal{S}; \boldsymbol{C}^N) \leq \frac{1}{\binom{p}{k}}\sum_{\mathcal{S}} \mathrm{KL}(P_{\mathcal{S}}^{\otimes N} \| P_0^{\otimes N}) = N\,\mathrm{KL}(P_{\mathcal{S}} \| P_0)$ for any fixed $\mathcal{S}$ by symmetry. Given $\mathcal{S}$, the coordinates of $\boldsymbol{c}$ are independent, equal in law to $\mathcal{N}(0, v_1)$ on $\mathcal{S}$ and $\mathcal{N}(0, v_0)$ off $\mathcal{S}$, and the reference has all coordinates $\mathcal{N}(0, v_0)$; hence $\mathrm{KL}(P_{\mathcal{S}} \| P_0) = k\,\mathrm{KL}(\mathcal{N}(0, v_1) \| \mathcal{N}(0, v_0)) = \frac{k}{2}\big(\frac{v_1}{v_0} - 1 - \log\frac{v_1}{v_0}\big) = k\,\mathrm{KL}_1(\rho)$. Rearranging Fano gives (1); the inequality $\mathrm{KL}_1 \leq \rho^2/4$ is $\log(1+\rho) \geq \rho - \rho^2/2$. □

Theorem S1.3 (Signal-agnostic Fano converse)**.** *Keep the noise Gaussian, but let the active triangle signals $y_{\tau,t}$ be i.i.d. from an* arbitrary *zero-mean law with variance σ_c^2. Then any estimator with error at most ε requires*

$$N \;\geq\; \frac{(1-\varepsilon)\log\binom{p}{k} - \log 2}{\frac{p}{2}\,\log\!\big(1 + \rho k/p\big)}. \tag{2}$$

Proof. Lemma S1.1 applies verbatim (its proof used only independence and Gaussianity of the *noise*). With conditionally i.i.d. snapshots, $I(\mathcal{S}; \boldsymbol{C}^N) \leq N\,I(\mathcal{S}; \boldsymbol{c}_1)$, and with conditionally independent coordinates, subadditivity of entropy gives $I(\mathcal{S}; \boldsymbol{c}) \leq \sum_\tau \big[h(c_\tau) - h(c_\tau \mid \mathcal{S})\big]$. Under the uniform prior each coordinate is active with probability k/p, so $\mathrm{Var}(c_\tau) = v_0(1 + \rho k/p)$ and the Gaussian maximum-entropy bound gives $h(c_\tau) \leq \frac{1}{2}\log(2\pi e\, v_0(1 + \rho k/p))$. Conditionally on $\mathcal{S}$, c_τ is an independent signal plus $\mathcal{N}(0, v_0)$ noise, so $h(c_\tau \mid \mathcal{S}) \geq \frac{1}{2}\log(2\pi e\, v_0)$ (adding an independent variable cannot decrease entropy). Summing over the p coordinates yields $I(\mathcal{S}; \boldsymbol{c}) \leq \frac{p}{2}\log(1 + \rho k/p)$ and Fano concludes. □

Remark S1.4 (How the two converses compare)**.** *Both bounds are valid for Gaussian signals, so one may take their pointwise maximum. For sparse supports ($k \ll p$) and small ρ the Gaussian bound is stronger by a factor $\asymp 1/\rho$ (its denominator is $k\mathrm{KL}_1 \asymp k\rho^2/4$ versus $\asymp k\rho/2$). In the dense regime ($k \asymp p$) the agnostic budget grows only logarithmically in ρ, so it* overtakes *the Gaussian-KL bound at high SNR (Fig. 1B). Figure 1A shows the* $\log p$ *effect: at the scale of the $p = 8$ planted-complex setting the joint-recovery floor sits well below the per-triangle Chernoff floor, while at $p = 10^4$ it narrows the gap to a $\approx 2\times$ factor (the curves are not directly comparable in level: Fano is normalized at error $1/2$, the Chernoff floor at $\delta = 0.05$).*

Proposition S1.5 (Sub-Gaussian achievability)**.** *Suppose the curl-coordinate noise and signals are sub-Gaussian with ψ_2-norms bounded by K times their standard deviations. Then the energy detector with threshold $\gamma = v_0(1 + \rho/2)$ recovers $\mathcal{S}$ exactly with probability $\geq 1 - \delta$ once*

$$N \;\geq\; \frac{C\,K^4}{\min(\rho, \rho^2)/(1+\rho)^2}\,\log\frac{2p}{\delta},$$

for a universal constant C: the same $\log p/\rho^2$ scaling as the Gaussian theory, with constants degraded by K^4.

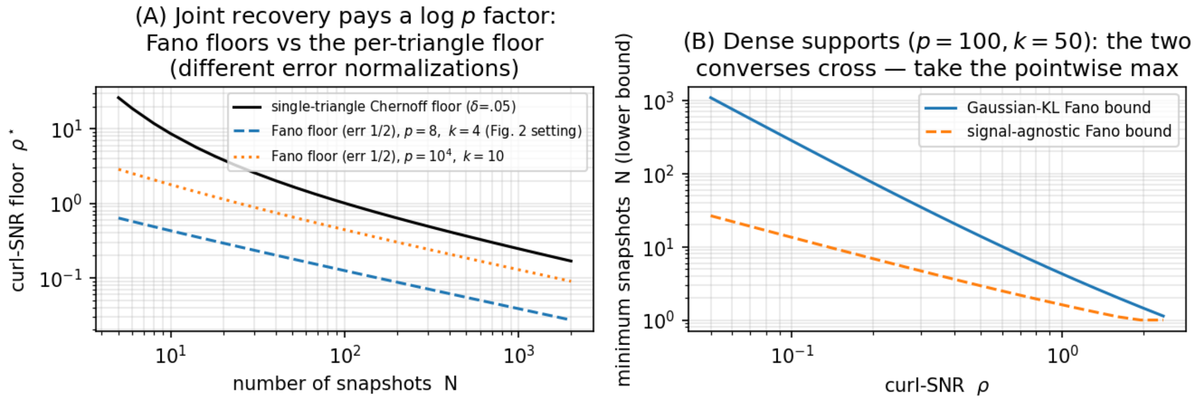

Figure 1: (A) Fano floors (Thm. S1.2, inverted in ρ at error 1/2) against the single-triangle Chernoff floor of the main paper. (B) Dense-support regime: the Gaussian-KL and signal-agnostic bounds cross. Script: `experiments/run_fano.py`.

Proof sketch. Each $c_{\tau,t}^2 - \mathbb{E}c_{\tau,t}^2$ is sub-exponential with ψ_1-norm $\lesssim K^2 v_i$. Bernstein's inequality [1, Thm. 2.8.1] gives, for the empirical energy $\hat{E}_\tau = \frac{1}{N}\sum_t c_{\tau,t}^2$,

$$\Pr\big[|\hat{E}_\tau - v_i| \geq t\big] \;\leq\; 2\exp\Big(-cN\min\big(t^2/(K^4 v_i^2),\; t/(K^2 v_i)\big)\Big).$$

The two hypotheses are separated by $v_1 - v_0 = v_0\rho$; taking $t = v_0\rho/2$ and a union bound over the p candidates yields the claim. $\square$

S2 Partial edge observation

Suppose only a subset $\mathcal{E}_{\text{obs}} \subseteq \mathcal{E}$ of edges is observed — each edge independently with probability q (Bernoulli sampling). Write $\boldsymbol{f}_t^{\text{obs}}$ for the observed sub-vector.

Proposition S2.1 (Observability for curl statistics)**.** *The curl statistic $c_\tau = \boldsymbol{B}_2[:,\tau]^\top \boldsymbol{f}$ of candidate τ is computable from $\boldsymbol{f}^{\text{obs}}$ iff* all three *edges of τ are observed. Under Bernoulli(q) sampling each triangle is observable with probability q^3, independently across edge-disjoint candidates.*

An unobservable *active* triangle is an automatic miss; an unobservable *inactive* one is an automatic correct rejection (the detector defaults to "inactive"). Combining with the exact per-triangle marginal laws (main paper, Prop. 2) gives a closed form on edge-disjoint geometries:

Corollary S2.2 (Exact recovery under edge sampling)**.** *With edge-disjoint candidates, the centered whitened detector at budget N achieves*

$$\Pr[\hat{\mathcal{S}} = \mathcal{S}] = \big[q^3\,(1 - P_{\text{miss}})\big]^k\;\big[1 - q^3 P_{\text{fa}}\big]^{p-k}, \tag{3}$$

where $P_{\text{miss}}, P_{\text{fa}}$ are the χ^2_{N-1} error probabilities of the fully observed test.

Proof. Independence of survival (Prop. S2.1) and of the whitened scores ($\boldsymbol{G} = 3\boldsymbol{I}$). An active triangle is recovered iff it survives (prob. q^3) *and* is detected ($1 - P_{\text{miss}}$); an inactive one is mislabeled iff it survives *and* false-alarms ($q^3 P_{\text{fa}}$). $\square$

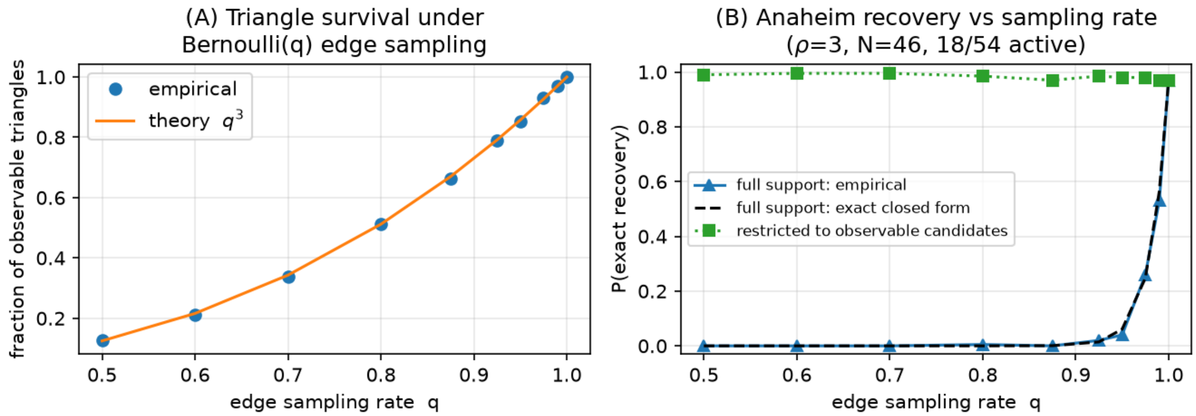

Figure 2: Partial edge observation on Anaheim (real geometry and equilibrium flow background, $\rho = 3$, $N = 46$, 18/54 active). (A) Triangle survival follows q^3. (B) Full-support recovery collapses along the exact closed form (3); recovery restricted to observable candidates stays ≈ 1. Script: `experiments/run_partial_sampling.py`.

The q^{3k} factor is brutal: partial sampling multiplies the fully observed recovery probability by $q^{3k} = q^{54}$ at $k = 18$ active triangles — a factor ≈ 0.06 at $q = 0.95$ and ≈ 0.003 at $q = 0.9$ — and the Monte-Carlo cliff in Fig. 2B follows the closed form within sampling error. The restricted curve (recovery on the observable candidates only) stays at ≈ 1 for all q: the bottleneck is purely coverage, not detection.

Remark S2.3 (Scope)**.** *Proposition S2.1 is an* operational *statement about detectors built on curl statistics — the class this paper analyzes. A triangle with two observed edges still injects signal into those edges' flows, but that signal is entangled with the unknown gradient nuisance once the exact curl cancellation is unavailable; whether (and at what cost) it can be exploited, e.g. by marginalizing the gradient over the observed sub-graph, is an open question we do not claim to settle. Low-rank flow completion under sampling (cf. graph sampling theory [2]) is the natural attack.*

S3 Plug-in variance estimation and the adaptive detector

The paper's detectors take (σ_c, σ_n) as known. We now estimate both from the same data. Let $u_\tau = \mathrm{score}_\tau/(\boldsymbol{G}^{-1})_{\tau\tau}$ denote the normalized whitened scores, so that for inactive τ, $N u_\tau/\sigma_n^2 \sim \chi^2_d$ ($d = N$ or $N-1$ if centered), while active scores are stochastically larger.

Estimators.

$$\hat{\sigma}_n^2 = \frac{N\,\mathrm{med}_\tau(u_\tau)}{\mathrm{med}(\chi^2_d)}, \qquad \text{(median, robust to} < 50\% \text{ contamination)} \tag{4}$$

$$\hat{\sigma}_c^2 = \left(\tfrac{1}{|\hat{\mathcal{A}}|}\textstyle\sum_{\tau\in\hat{\mathcal{A}}}\left[\mathrm{score}_\tau - \hat{v}_{0,\tau}\right]\right)_+, \qquad \hat{\mathcal{A}} = \text{Bonferroni screen at level } \alpha, \tag{5}$$

followed by one refinement pass: after a first detection, σ_n is re-fit by the mean of the *detected* off-support scores and σ_c by the on-support excess, and the detector is re-run once. Because the refit set is selected by the same data, the refit is conditionally biased (missed active triangles inflate $\hat{\sigma}_n$ — measured $\approx +5\%$ at $N = 10$ on the strip benchmark, vanishing for $N \geq 20$); its accuracy is established empirically (Fig. 3A), not by an unbiasedness argument, and Prop. S3.2 applies conditionally on the deterministic error bound ϵ.

Lemma S3.1 (Median consistency envelope)**.** *Assume the candidates are pairwise edge-disjoint, so that the inactive normalized scores are i.i.d. $\sigma_n^2 \chi_d^2 / N$ (the DKW step below requires independence; for correlated scores the envelope is validated only empirically). Let a fraction $\pi < 1/2$ of the p candidates be active and let $\epsilon_p = \sqrt{\log(2/\delta)/(2(1-\pi)p)}$ satisfy $\pi + \epsilon_p < 1/2$. Then with probability at least $1 - \delta$,*

$$q_d\Big(\tfrac{1/2-\pi-\epsilon_p}{1-\pi}\Big) \;\leq\; \frac{N\,\mathrm{med}_\tau(u_\tau)}{\sigma_n^2} \;\leq\; q_d\Big(\tfrac{1/2}{1-\pi} + \tfrac{\epsilon_p}{1-\pi}\Big),$$

where $q_d(\cdot)$ is the χ_d^2 quantile function. Consequently $\hat{\sigma}_n^2/\sigma_n^2 \in [q_d(\frac{1/2-\pi-\epsilon_p}{1-\pi}), q_d(\frac{1/2+\epsilon_p}{1-\pi})]/q_d(1/2)$ — an explicit, numerically computable envelope that shrinks to $\{1\}$ as $p, d \to \infty$ whenever $\pi < 1/2$.

Proof. Active scores stochastically dominate inactive ones, so the sample median of all p scores is sandwiched between order statistics of the $(1-\pi)p$ inactive scores: at least a $(1/2-\pi)/(1-\pi)$-fraction and at most a $(1/2)/(1-\pi)$-fraction of the inactive sample lies below it. The Dvoretzky–Kiefer–Wolfowitz inequality applied to the inactive empirical CDF converts these fractions into population χ_d^2 quantiles up to ϵ_p, with probability $1 - \delta$. □

Remark S3.2 (Plug-in perturbation — empirical observation)**.** *We do* not *prove a uniform explicit-constant bound on the adaptive detector's excess error. Heuristically, with plug-in variances $\hat{v}_{i,\tau} = (1 + e_{i,\tau})v_{i,\tau}$, $|e_{i,\tau}| \leq \epsilon < 1/2$, the Bayes threshold $\gamma(v_0, v_1)$ is smooth (main paper, Prop. 2) so a first-order expansion suggests each per-triangle error probability changes by $O(\epsilon)$ and the exact-recovery probability by $O(p\epsilon)$ (union bound); but the constant depends on the χ_d^2 density at the (shifting) threshold and on the $\hat{v}$ concentration, and we have not established a* uniform *Lipschitz constant over the operating range. We therefore report the effect empirically only. Establishing a non-asymptotic joint selection-plus-estimation bound with explicit constants is left open.*

Empirically (Fig. 3) both variance estimates contract at the $1/\sqrt{N}$ rate and the fully adaptive detector's recovery curve is indistinguishable from the known-parameter detector for $N \gtrsim 30$ on the confusable strip benchmark; the residual gap at $N \leq 20$ (at most 0.06 after refinement) is the price of estimating a variance from nine medians. These are measurements, not a theorem.

S4 Excitation-dependent identifiability: proofs

This section proves the main paper's Lemma 1 (lifted spark), Theorem 1 (three excitation regimes), Theorem 2 (separations and exponents), and Theorem 3 (NNLS consistency and failure bound), and records the exhaustive numerical verifications. Here the candidate set is arbitrary (edge-sharing allowed, $\boldsymbol{B}_2$ possibly rank-deficient); $\rho_2 = \sigma_c^2/\sigma_n^2$ denotes the second-order SNR (the per-triangle curl-SNR is $\rho = 3\rho_2$).

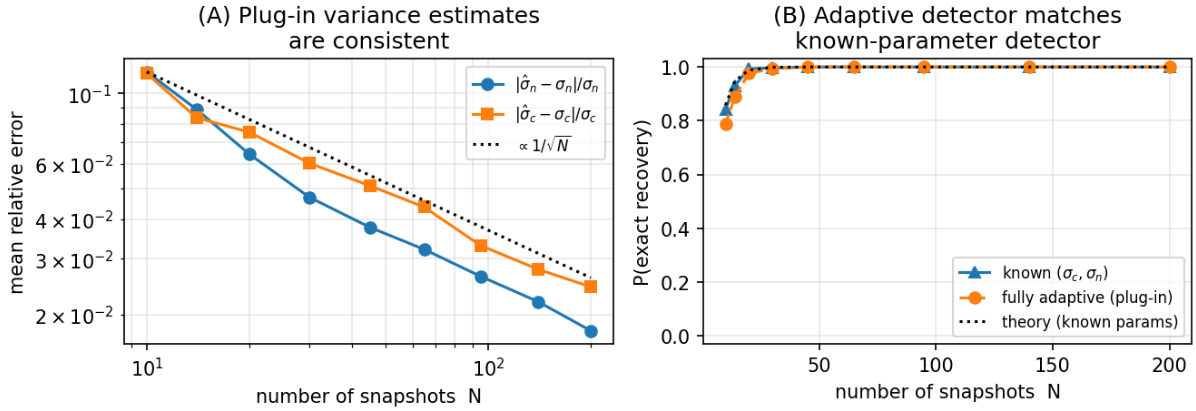

Figure 3: Plug-in estimation on the edge-sharing strip benchmark ($\rho = 8$, 4/9 active; cf. main Prop. 1). (A) Consistency of (4)–(5) (after refinement). (B) The fully adaptive detector matches the known-parameter detector. Script: `experiments/run_plugin.py`.

S4.1 The lifted spark lemma

Lemma S4.1 (Lifted spark)**.** *Let $\mathcal{T}$ be any set of distinct 3-cliques of a simple graph, with signatures $\boldsymbol{b}_\tau \in \{0, \pm 1\}^{|\mathcal{E}|}$ (columns of $\boldsymbol{B}_2$). The matrices $\{\boldsymbol{b}_\tau \boldsymbol{b}_\tau^\top\}_{\tau \in \mathcal{T}}$ are linearly independent.*

Proof. Fix $\tau = (i, j, k)$ and its edge pair $e = (i, j)$, $e' = (j, k)$. A pair of distinct edges determines at most one triangle containing both (two edges sharing a vertex fix the third vertex; disjoint edges lie in no common triangle). Hence in $\boldsymbol{A}(\boldsymbol{w}) = \sum_\sigma w_\sigma \boldsymbol{b}_\sigma \boldsymbol{b}_\sigma^\top$, the (e, e') entry equals $w_\tau\, \boldsymbol{b}_\tau(e) \boldsymbol{b}_\tau(e') = \pm w_\tau$: no other candidate contributes. $\boldsymbol{A}(\boldsymbol{w}) = \boldsymbol{0}$ therefore forces $w_\tau = 0$ for every τ. □

S4.2 The three excitation regimes (main Theorem 1)

With $\boldsymbol{z} = \boldsymbol{Q}^\top \boldsymbol{f}$ ($\boldsymbol{Q}$ an orthonormal basis of $\operatorname{im} \boldsymbol{B}_2$, $r = \operatorname{rank} \boldsymbol{B}_2$, $\boldsymbol{U} = \boldsymbol{Q}^\top \boldsymbol{B}_2$), the gradient term is annihilated unconditionally ($\boldsymbol{B}_2^\top \boldsymbol{B}_1^\top = \boldsymbol{0}$) and the harmonic term by the candidate-orthogonality assumption $\boldsymbol{h} \in \ker \boldsymbol{B}_2^\top$, so $\boldsymbol{\Sigma}_z(\mathcal{S}, \boldsymbol{\Gamma}) = \sigma_n^2 \boldsymbol{I}_r + \boldsymbol{U}_\mathcal{S} \boldsymbol{\Gamma} \boldsymbol{U}_\mathcal{S}^\top$; since $\boldsymbol{b}_\tau \in \operatorname{im} \boldsymbol{B}_2$, the compression loses nothing ($\boldsymbol{u}_\sigma^\top \boldsymbol{u}_\tau = G_{\sigma\tau}$).

Proposition S4.2 ((a) positive diagonal)**.** *If $\boldsymbol{\Gamma}_\mathcal{S} = \operatorname{diag}(\gamma_\tau) > 0$, then $\boldsymbol{\Sigma}_z - \sigma_n^2 \boldsymbol{I} = \sum_\tau w_\tau \boldsymbol{u}_\tau \boldsymbol{u}_\tau^\top$ with $w_\tau = \gamma_\tau \mathbf{1}\{\tau \in \mathcal{S}\}$, and by Lemma S4.1 the coefficients w (hence $\mathcal{S}$ and the weights) are uniquely determined — at any rank deficiency of $\boldsymbol{B}_2$. On K_n with σ_n* unknown, *exactly one direction is ambiguous: $\sum_{\tau \in \mathcal{T}} \boldsymbol{u}_\tau \boldsymbol{u}_\tau^\top = n \boldsymbol{I}_r$ (the up-Laplacian of the full complex acts as n on the curl space), so a uniform weight shift trades off against the noise floor; this is verified numerically for K_5–K_7.*

Proposition S4.3 ((b) Arbitrary PSD)**.** *For each support the achievable-covariance set is*

$$\mathcal{C}_\mathcal{S} = \{\boldsymbol{U}_\mathcal{S} \boldsymbol{\Gamma} \boldsymbol{U}_\mathcal{S}^\top : \boldsymbol{\Gamma} \succeq 0\} = \{\boldsymbol{M} \succeq 0 : \operatorname{im} \boldsymbol{M} \subseteq \operatorname{im} \boldsymbol{U}_\mathcal{S}\}.$$

Consequently, given the population *covariance $\boldsymbol{\Sigma}_z$ (equivalently $\boldsymbol{M} = \boldsymbol{\Sigma}_z - \sigma_n^2 \boldsymbol{I}$), the set of supports consistent with it is exactly $\{\mathcal{S} : \operatorname{im} \boldsymbol{M} \subseteq \operatorname{im} \boldsymbol{U}_\mathcal{S}\}$. Hence:* (i) *the realized range $\mathcal{R} = \operatorname{im} \boldsymbol{M} = \operatorname{im}(\boldsymbol{U}_\mathcal{S} \boldsymbol{\Gamma}^{1/2}) \subseteq \operatorname{im} \boldsymbol{U}_\mathcal{S}$ is always identifiable;* (ii) *the candidate image $\operatorname{im} \boldsymbol{B}_{2,\mathcal{S}}$ is* not — *any support with a larger image and a suitable* $\boldsymbol{\Gamma}$ *realizes the same* $\boldsymbol{M}$*;* (iii) *a full-rank* $\boldsymbol{\Gamma} \succ 0$ *gives* $\mathcal{R} = \operatorname{im} \boldsymbol{U}_{\mathcal{S}}$ *(the converse only when* $\boldsymbol{U}_{\mathcal{S}}$ *is injective).*

Proof. Set identity: $(\subseteq)$ $\boldsymbol{U}_{\mathcal{S}} \boldsymbol{\Gamma} \boldsymbol{U}_{\mathcal{S}}^\top \succeq 0$ has range in $\operatorname{im} \boldsymbol{U}_{\mathcal{S}}$; $(\supseteq)$ for PSD $\boldsymbol{M}$ with $\operatorname{im} \boldsymbol{M} \subseteq \operatorname{im} \boldsymbol{U}_{\mathcal{S}}$ set $\boldsymbol{\Gamma} = \boldsymbol{U}_{\mathcal{S}}^+ \boldsymbol{M} (\boldsymbol{U}_{\mathcal{S}}^+)^\top \succeq 0$, giving $\boldsymbol{U}_{\mathcal{S}} \boldsymbol{\Gamma} \boldsymbol{U}_{\mathcal{S}}^\top = \boldsymbol{P}_{\operatorname{im} \boldsymbol{U}_{\mathcal{S}}} \boldsymbol{M} \boldsymbol{P}_{\operatorname{im} \boldsymbol{U}_{\mathcal{S}}} = \boldsymbol{M}$. The support statement is immediate since $\boldsymbol{M} \in \mathcal{C}_{\mathcal{S}} \Leftrightarrow \operatorname{im} \boldsymbol{M} \subseteq \operatorname{im} \boldsymbol{U}_{\mathcal{S}}$. Range: $\boldsymbol{\Gamma} \succ 0 \Rightarrow \boldsymbol{\Gamma}^{1/2}$ invertible $\Rightarrow \mathcal{R} = \operatorname{im} \boldsymbol{U}_{\mathcal{S}}$; if $\boldsymbol{U}_{\mathcal{S}}$ is injective, $\dim \mathcal{R} = \operatorname{rank} \boldsymbol{\Gamma}$, so equality forces $\boldsymbol{\Gamma} \succ 0$. □

Remark S4.4 (Strong witness: strictly positive diagonal is not enough)**.** *Positive per-triangle variance does not rescue identifiability; regime (a) needs* $\boldsymbol{\Gamma}$ diagonal*. On* K_4 *with all four faces,* $\boldsymbol{U}_{\mathcal{S}}$ *has a one-dimensional kernel spanned by* $\boldsymbol{c} = (1, -1, 1, -1)^\top$ *(the tetrahedral relation). Take* $\boldsymbol{v} = \boldsymbol{e}_1 - \frac{1}{4}\boldsymbol{c} = (.75, .25, -.25, .25)^\top$ *(all entries nonzero) and* $\boldsymbol{\Gamma}' = \boldsymbol{v}\boldsymbol{v}^\top$*: then* $\boldsymbol{\Gamma}'$ *has* strictly positive diagonal*, rank* 1*, and* $\boldsymbol{U}_{\mathcal{S}} \boldsymbol{\Gamma}' \boldsymbol{U}_{\mathcal{S}}^\top = (\boldsymbol{U}_{\mathcal{S}} \boldsymbol{v})(\boldsymbol{U}_{\mathcal{S}} \boldsymbol{v})^\top = \boldsymbol{u}_1 \boldsymbol{u}_1^\top$ *(since* $\boldsymbol{U}_{\mathcal{S}} \boldsymbol{c} = \boldsymbol{0}$*) — identical to the singleton* $\{\tau_1\}$ *with unit variance, although* $\dim \operatorname{im} \boldsymbol{B}_{2,\mathcal{S}} = 3 \neq 1$*. The obstruction is the off-diagonal* correlation *of* $\boldsymbol{\Gamma}'$ *(*≈ 0.19*), not any zero variance (the degenerate* $\boldsymbol{\Gamma} = \operatorname{diag}(1, 0)$ *example needs an inactive triangle; this one does not). Verified in* `tests/test_excitation.py`*.*

Proposition S4.5 ((c) projector excitation)**.** *For* $\boldsymbol{\Gamma}_{\mathcal{S}} = \sigma_c^2 (\boldsymbol{B}_{2,\mathcal{S}}^\top \boldsymbol{B}_{2,\mathcal{S}})^+$*,* $\boldsymbol{U}_{\mathcal{S}} \boldsymbol{\Gamma}_{\mathcal{S}} \boldsymbol{U}_{\mathcal{S}}^\top = \sigma_c^2 \boldsymbol{P}_{\operatorname{im} \boldsymbol{U}_{\mathcal{S}}}$ *exactly (SVD:* $\boldsymbol{A}(\boldsymbol{A}^\top \boldsymbol{A})^+ \boldsymbol{A}^\top = \boldsymbol{P}_{\operatorname{im} \boldsymbol{A}}$*). Supports with equal curl image therefore induce identical snapshot distributions — indistinguishable at every SNR and* N*.*

Remark S4.6 (Deterministic boundary)**.** *Unknown deterministic signals are the rank-one degenerate boundary of regime (b): for equal-image supports the mean families* $\{\boldsymbol{B}_{2,\mathcal{S}} \boldsymbol{y}\} = \{\boldsymbol{B}_{2,\mathcal{S}'} \boldsymbol{y}'\}$ *coincide per snapshot, so no test distinguishes them at any SNR and any* N*. Tetrahedral confusers realize equal image: the four faces satisfy* $\boldsymbol{b}_{012} - \boldsymbol{b}_{013} + \boldsymbol{b}_{023} - \boldsymbol{b}_{123} = \boldsymbol{0}$*, so any three faces span the same* 3*-space as any other three; on* K_n *there are* $\binom{n}{4}$ *tetrahedra.*

S4.3 The price: exact separations and Chernoff exponents

Lemma S4.7 (Swap separation)**.** *Let* $\mathcal{S}' = \mathcal{S} \setminus \{\tau_a\} \cup \{\tau_b\}$ *with* τ_a, τ_b *faces of a common tetrahedron. Then* $\boldsymbol{\Sigma}_z(\mathcal{S}) - \boldsymbol{\Sigma}_z(\mathcal{S}') = \sigma_c^2 (\boldsymbol{u}_a \boldsymbol{u}_a^\top - \boldsymbol{u}_b \boldsymbol{u}_b^\top)$ *and*

$$\|\boldsymbol{u}_a \boldsymbol{u}_a^\top - \boldsymbol{u}_b \boldsymbol{u}_b^\top\|_F^2 = \|\boldsymbol{u}_a\|^4 + \|\boldsymbol{u}_b\|^4 - 2(\boldsymbol{u}_a^\top \boldsymbol{u}_b)^2 = 9 + 9 - 2 = 16,$$

independent of n*, of* $|\mathcal{S}|$*, and of the hosting tetrahedron (any two faces share exactly one edge, so* $\boldsymbol{u}_a^\top \boldsymbol{u}_b = G_{ab} = \pm 1$*).*

Lemma S4.8 (Subset separation — the unrestricted worst case)**.** *Let* $\mathcal{S}' = \mathcal{S} \cup \{\tau_4\}$ *where* τ_4 *is the fourth face of a tetrahedron whose other three faces lie in* $\mathcal{S}$*. Then* $\operatorname{im} \boldsymbol{B}_{2,\mathcal{S}} = \operatorname{im} \boldsymbol{B}_{2,\mathcal{S}'}$ *(the face signatures are linearly dependent) while* $\boldsymbol{\Sigma}_z(\mathcal{S}') - \boldsymbol{\Sigma}_z(\mathcal{S}) = \sigma_c^2 \boldsymbol{u}_4 \boldsymbol{u}_4^\top$ *with separation* $\|\boldsymbol{u}_4 \boldsymbol{u}_4^\top\|_F^2 = \|\boldsymbol{u}_4\|^4 = 9$*. Exhaustively over ALL* 2^{10} *supports of* K_5 *(*`test_exhaustive_k5_confuser_separations`*:* 45,000+ *equal-image distinct pairs),* 9 *is the* unrestricted minimum *separation, while restricted to pairs of* equal cardinality *the minimum is* 16*, attained exactly by the swaps of Lemma S4.7. (An earlier revision of this project wrongly claimed* 16 *as the unrestricted minimum; the equal-cardinality scoping is essential.)*

Proposition S4.9 (Chernoff exponents; rank deficiency is free at the exponent level)**.** *Let* C_G *be the Chernoff information between* $\mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}_z(\mathcal{S}))$ *and* $\mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}_z(\mathcal{S}'))$*, and write the DIMENSIONLESS signature Gram* $\boldsymbol{D}_{\mathcal{S}} = \sum_{\tau \in \mathcal{S}} \boldsymbol{u}_\tau \boldsymbol{u}_\tau^\top$ *(so* $\boldsymbol{M}_{\mathcal{S}} = \sigma_c^2 \boldsymbol{D}_{\mathcal{S}}$*). As* $\rho_2 \to 0$*,*

$$C_G = \tfrac{1}{16}\, \rho_2^2 \, \|\Delta \boldsymbol{D}\|_F^2 \, (1 + o(1)) = \tfrac{1}{16\sigma_n^4} \, \|\Delta \boldsymbol{M}\|_F^2 \, (1 + o(1)) = \begin{cases} \rho_2^2 (1 + o(1)) & \textit{swap}, \\ \frac{9}{16} \rho_2^2 (1 + o(1)) & \textit{subset}, \end{cases}$$

by Lemmas S4.7–S4.8; the two forms coincide since $\Delta\boldsymbol{M} = \sigma_c^2 \Delta\boldsymbol{D}$ *(do* not *write* $\rho_2^2\|\Delta\boldsymbol{M}\|_F^2$*, which double-counts* σ_c^4*). the optimal error satisfies* $P_{\mathrm{err}} \le e^{-NC_G}$ *with matching exponent, and* $N^\star(\delta) \sim \log(1/\delta)/C_G$*. Moreover the subset exponent* equals *the isolated-triangle detection exponent:* $C(\rho) = \rho^2/16$ *at* $\rho = 3\rho_2$ *is* $\frac{9}{16}\rho_2^2$ *(numerically checked to* 3×10^{-5} *relative at* $\rho_2 = 10^{-5}$*, `test_subset_confuser_exponent_equals_isolated_triangle_exponent`). Hence the hardest equal-image decision costs, at the exponent level, exactly one isolated-triangle detection: geometric rank deficiency imposes no exponent penalty, and only the Fano log-multiplicity of the confuser family reflects the geometry.*

Proof sketch. Write $\boldsymbol{E} = \boldsymbol{\Sigma}_0^{-1/2}(\boldsymbol{\Sigma}_1 - \boldsymbol{\Sigma}_0)\boldsymbol{\Sigma}_0^{-1/2}$. The Chernoff exponent between equal-mean Gaussians is $\max_s \frac{1}{2}[\log\det((1-s)\boldsymbol{W}_0 + s\boldsymbol{W}_1) - (1-s)\log\det\boldsymbol{W}_0 - s\log\det\boldsymbol{W}_1]$; a second-order expansion in $\boldsymbol{E}$ around $\boldsymbol{E} = \boldsymbol{0}$ is maximized at $s = 1/2$ (Bhattacharyya) with value $\frac{1}{16}\operatorname{tr}\boldsymbol{E}^2 + O(\|\boldsymbol{E}\|^3)$. Here $\boldsymbol{E} = \rho_2(\boldsymbol{u}_a\boldsymbol{u}_a^\top - \boldsymbol{u}_b\boldsymbol{u}_b^\top) + O(\rho_2^2)$ (the shared atoms of $\mathcal{S}\cap\mathcal{S}'$ perturb $\boldsymbol{\Sigma}_0^{-1/2}$ only at $O(\rho_2)$, entering C_G at $O(\rho_2^3)$), so $\operatorname{tr}\boldsymbol{E}^2 = \rho_2^2 \cdot 16\,(1 + O(\rho_2))$ by Lemma S4.7. The bound $P_{\mathrm{err}} \le e^{-NC_G}$ is the Chernoff bound; exponent tightness is classical for i.i.d. observations. □

Proposition S4.10 (Fano over the confuser family)**.** *On* K_n*, consider the* $M = 4\binom{n}{4}$ *hypotheses obtained by choosing a tetrahedron and leaving one of its four faces out of an otherwise-active tetrad. Any estimator with error* $\le \epsilon$ *over the uniform prior needs*

$$N \;\ge\; \frac{(1-\epsilon)\log\big(4\binom{n}{4}\big) - \log 2}{\max_{pairs}\mathrm{KL}\big(\boldsymbol{\Sigma}_z\|\boldsymbol{\Sigma}_z'\big)} \;\asymp\; \frac{\log n}{\rho_2^2},$$

i.e. full support recovery over the confuser family pays a $\log n$ *factor on top of the pairwise budget.*

S4.4 Global analytic separations (main Theorem 2)

Theorem S4.11 (Separations on every clique complex)**.** *Let* $\mathcal{T}$ *be any candidate set of distinct 3-cliques of a simple graph and let* $\mathcal{S} \ne \mathcal{S}'$ *be binary supports with equal curl image, isotropic excitation* $\sigma_c^2\boldsymbol{I}$*. Then* $\|\boldsymbol{M}_{\mathcal{S}} - \boldsymbol{M}_{\mathcal{S}'}\|_F^2 \;\ge\; 9\sigma_c^4$*, with equality iff* $\mathcal{S},\mathcal{S}'$ *differ in exactly one (linearly dependent) face; and* $\ge 16\sigma_c^4$ *if additionally* $|\mathcal{S}| = |\mathcal{S}'|$*, with equality iff the difference is a single edge-sharing swap.*

Proof. Write $\boldsymbol{M}_{\mathcal{S}} - \boldsymbol{M}_{\mathcal{S}'} = \sigma_c^2\sum_\tau c_\tau \boldsymbol{b}_\tau\boldsymbol{b}_\tau^\top$ with $c_\tau \in \{-1, 0, +1\}$, $\boldsymbol{c} \ne \boldsymbol{0}$. Then

$$\Big\|\sum_\tau c_\tau \boldsymbol{b}_\tau\boldsymbol{b}_\tau^\top\Big\|_F^2 = \sum_{\sigma,\tau} c_\sigma c_\tau G_{\sigma\tau}^2 = \boldsymbol{c}^\top(9\boldsymbol{I} + \boldsymbol{A})\boldsymbol{c},$$

since $G_{\tau\tau}^2 = 9$ and $G_{\sigma\tau}^2 \in \{0,1\}$ equals the share-an-edge adjacency $\boldsymbol{A}$. Two triangles share an edge iff their vertex triples intersect in two elements, so $\boldsymbol{A}$ is the adjacency of an induced subgraph of the Johnson graph $J(n,3)$ (n = number of vertices); the Johnson spectrum $\lambda_j = (3-j)(n-3-j)-j$ gives $\lambda_{\min} \ge -3$, and Cauchy interlacing transfers this to every induced subgraph (for $n < 6$, direct computation gives $\lambda_{\min} \in \{-1,-2\}$). Hence sep $\ge (9-3)\|\boldsymbol{c}\|^2 = 6\|\boldsymbol{c}\|^2$. If $\|\boldsymbol{c}\|^2 = 1$ (single-face difference) then $\boldsymbol{c}^\top\boldsymbol{A}\boldsymbol{c} = 0$ and sep $= 9$ exactly; equal image with a single-face difference requires that face's signature to lie in the span of the shared columns (e.g. a hosted tetrahedron's fourth face). If $\|\boldsymbol{c}\|^2 \ge 2$ then sep $\ge 12 > 9$, so 9 is the unrestricted minimum. Under $|\mathcal{S}| = |\mathcal{S}'|$, $\sum_\tau c_\tau = 0$ forces $\|\boldsymbol{c}\|^2$ even and ≥ 2; for $\|\boldsymbol{c}\|^2 = 2$ ($\boldsymbol{c} = \boldsymbol{e}_a - \boldsymbol{e}_b$), sep $= 18 - 2G_{ab}^2 \ge 16$ with equality iff the two faces share an edge (tetrahedral swap); for $\|\boldsymbol{c}\|^2 \ge 4$, sep $\ge 24 > 16$. □

The Chernoff exponents then follow from the Bhattacharyya expansion (Prop. S4.9), for *fixed graph and support pair* as $\rho_2 \to 0$; in particular the unrestricted worst case (subset confuser, separation 9) has exponent $\frac{9}{16}\rho_2^2(1+o(1))$, equal to the isolated-triangle detection exponent.

S4.5 NNLS consistency and failure bound (main Theorem 3)

Proof of main Theorem 3. Let $\boldsymbol{A} = [\mathrm{vec}(\boldsymbol{u}_\tau \boldsymbol{u}_\tau^\top)]_\tau$, $\boldsymbol{s} = \mathrm{vec}(\hat{\boldsymbol{\Sigma}}_z - \sigma_n^2 \boldsymbol{I})$, true coefficients $\boldsymbol{w}^\star \geq 0$ with $\boldsymbol{A}\boldsymbol{w}^\star = \mathrm{vec}(\boldsymbol{\Sigma}_z - \sigma_n^2 \boldsymbol{I})$, and $\boldsymbol{E} = \mathrm{vec}(\hat{\boldsymbol{\Sigma}}_z - \boldsymbol{\Sigma}_z)$. *(i) Deterministic perturbation.* Optimality of $\hat{\boldsymbol{w}}$ over the cone $\{\boldsymbol{w} \geq 0\} \ni \boldsymbol{w}^\star$ gives $\|\boldsymbol{A}\hat{\boldsymbol{w}} - \boldsymbol{s}\|^2 \leq \|\boldsymbol{A}\boldsymbol{w}^\star - \boldsymbol{s}\|^2 = \|\boldsymbol{E}\|^2$; expanding $\boldsymbol{A}\hat{\boldsymbol{w}} - \boldsymbol{s} = \boldsymbol{A}(\hat{\boldsymbol{w}} - \boldsymbol{w}^\star) - \boldsymbol{E}$ yields $\|\boldsymbol{A}(\hat{\boldsymbol{w}} - \boldsymbol{w}^\star)\|^2 \leq 2\langle \boldsymbol{A}(\hat{\boldsymbol{w}} - \boldsymbol{w}^\star), \boldsymbol{E}\rangle \leq 2\|\boldsymbol{A}(\hat{\boldsymbol{w}} - \boldsymbol{w}^\star)\|\|\boldsymbol{E}\|$, so $\|\boldsymbol{A}(\hat{\boldsymbol{w}} - \boldsymbol{w}^\star)\| \leq 2\|\boldsymbol{E}\|$ and $\|\hat{\boldsymbol{w}} - \boldsymbol{w}^\star\|_2 \leq 2\|\boldsymbol{E}\|/\sigma_{\min}(\boldsymbol{A})$, with $\sigma_{\min}(\boldsymbol{A}) > 0$ by Lemma S4.1. *(ii) Exact covariance-error moment.* For i.i.d. $\boldsymbol{z}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}_z)$ and $\hat{\boldsymbol{\Sigma}}_z = \frac{1}{N}\sum_t \boldsymbol{z}_t \boldsymbol{z}_t^\top$, Wishart second moments give $\mathrm{Var}(\hat{\Sigma}_{ij}) = (\Sigma_{ii}\Sigma_{jj} + \Sigma_{ij}^2)/N$, hence $\mathbb{E}\|\boldsymbol{E}\|^2 = \sum_{ij} \mathrm{Var}(\hat{\Sigma}_{ij}) = [(\mathrm{tr}\,\boldsymbol{\Sigma}_z)^2 + \|\boldsymbol{\Sigma}_z\|_F^2]/N$ exactly. *(iii) Markov.* Thresholding at $w_{\min}/2$ fails only if $\|\hat{\boldsymbol{w}} - \boldsymbol{w}^\star\|_\infty \geq w_{\min}/2$, so

$$\Pr[\hat{\mathcal{S}} \neq \mathcal{S}] \leq \Pr\left[\|\boldsymbol{E}\| \geq \frac{\sigma_{\min}(\boldsymbol{A})\, w_{\min}}{4}\right] \leq \frac{16\,\mathbb{E}\|\boldsymbol{E}\|^2}{\sigma_{\min}^2(\boldsymbol{A})\, w_{\min}^2},$$

which is the stated bound. Consistency: $\hat{\boldsymbol{\Sigma}}_z \to \boldsymbol{\Sigma}_z$ a.s. (LLN), so by (i) $\hat{\boldsymbol{w}} \to \boldsymbol{w}^\star$ a.s., and any threshold in $(0, w_{\min})$ recovers $\mathcal{S}$ eventually. Each step is separately Monte-Carlo verified (`test_excitation.py`): the moment identity to 6% relative at $N = 40$, zero violations of (i) in 220 trials (200 at N=30, 20 at N=2000), and the final bound upper-bounds empirical failure across an (N, ρ_2) grid. □

S4.6 Numerical verification (released tests)

`tests/test_second_order.py` certifies, deterministically or by Monte-Carlo: (i) Lemma S4.1 on K_4–K_9, strips, disjoint unions, and random clique complexes; (ii) injectivity of $\mathcal{S} \mapsto \boldsymbol{\Sigma}_z(\mathcal{S})$ *exhaustively* over all 2^4 supports of K_4; (iii) the deterministic replication of first-order confusers (residual $< 10^{-9}$); (iv) exhaustively over ALL 2^{10} supports of K_5 (45,000+ equal-image distinct pairs): unrestricted minimum separation = 9 (subset confusers, Lemma S4.8) and equal-cardinality minimum = 16 (swaps, Lemma S4.7); (v) swap C_G/ρ_2^2 within 6×10^{-4} of 1 at $\rho_2 = 10^{-4}$ (and within 5% at $\rho_2 \leq 10^{-2}$), subset C_G matching the isolated-triangle exponent to 3×10^{-5} relative at $\rho_2 = 10^{-5}$, and the Monte-Carlo error of the optimal test within the e^{-NC_G} envelope with empirical exponent decreasing onto C_G; (vi) exact greedy covariance-atom recovery on rank-deficient K_5. `tests/test_excitation.py` additionally checks every claim of the regime and estimator theory: weighted-diagonal identifiability and the K_n unknown-noise ambiguity direction ($\sum_\tau \boldsymbol{u}_\tau \boldsymbol{u}_\tau^\top = n\boldsymbol{I}_r$); PSD matching for equal-image supports (Prop. S4.3, residual $< 10^{-9}$, matched $\boldsymbol{\Gamma}'$ PSD); exact projector-excitation indistinguishability (Prop. S4.5); the separation identity $\boldsymbol{c}^\top(9\boldsymbol{I} + \boldsymbol{A})\boldsymbol{c}$ and $\lambda_{\min}(\boldsymbol{A}) \geq -3$ on K_4–K_7 and random clique complexes; the three steps of main Theorem 3's proof individually plus the assembled failure bound against empirical failure rates on an (N, ρ_2) grid; the population score-tie of the matched-subspace baseline and its collapse (with NNLS) to chance under projector excitation; and the monotone vanishing of the equal-image gap under the $\boldsymbol{\Gamma}_\alpha$ interpolation, exactly zero at $\alpha = 1$.

S5 The cyclone study: construction details

This section records the methodological details of the main paper's curl-based vortex-localization study (Fig. 3 there) beyond what fits in the page budget. Positioning: this is *not* filled-triangle topology recovery — the mesh has no equal-image confusers and weather has no latent boolean support; it grounds the sufficient statistic on genuinely unplanted rotational structure.

From winds to edge flows. ERA5 10 m winds (ARCO-ERA5 public archive, 0.703° equiangular grid, 6-hourly), Western North Pacific (0–45°N, 100–180°E), Aug–Sep 2020 (244 snapshots; the season contains 13 IBTrACS storms, incl. Bavi, Maysak, Haishen). Mesh nodes subsample every third grid point ($\approx 2.1°$); each grid cell splits into two right triangles; edge flow = trapezoidal line integral $\frac{1}{2}(\boldsymbol{V}_i + \boldsymbol{V}_j)\cdot(\Delta x, \Delta y)$ in the local equirectangular metric ($\Delta x = R\cos(\bar{\phi})\Delta\lambda$). The mesh is simply connected, so #triangles $= |\mathcal{E}| - |\mathcal{V}| + 1 = \dim\ker \boldsymbol{B}_1$ and $\boldsymbol{B}_2$ has full column rank (verified: 1554/1554) — the favorable-geometry regime (a) of the excitation hierarchy, with no equal-image confusers. By Stokes' theorem the curl statistic of a triangle approximates its circulation, i.e. area-integrated relative vorticity; a sign convention (sorted-vertex traversal vs. geometric orientation) is tracked explicitly in the code and is irrelevant to the energy detector.

Detection. Per 4-day window (16 snapshots), the headline statistic is the temporally centered curl-*energy* score of the main paper's Prop. 2, area^2-normalized to a common mean-vorticity scale; it is parameter-free (no thresholds are set — evaluation is by ROC ranking), and nothing is planted. The decorrelated (GLS, main Prop. 1) score is computed alongside and reported in the released JSON; its $\boldsymbol{G}^{-1}$ noise amplification hurts pure detection ranking on this near-regular full-rank mesh (internal AUC 0.773 vs. 0.914). Honest statement about motion: cyclones translate 3–6°/day, i.e. ∼6–11 triangle-widths over a 4-day window — the latent support is *not* static. Temporal centering removes the window-mean flow, so what the statistic detects is the *time-varying* circulation that a translating vortex deposits in every triangle it crosses (a perfectly static vortex would be removed by the centering); the i.i.d.-snapshot theory is thus a working model here, mapped through window-pooled ROC evaluation rather than through per-window support recovery. README honesty note 9 states the same.

Validation. Internal: relative vorticity $\zeta = \partial_x v - \partial_y u$ by centered differences on the FULL 0.7° grid (the detector only ever sees the 2.1° mesh flows); per-triangle truth = window-mean $|\zeta|$ averaged over grid points inside the triangle; ROC sweeps the detector score against truth binarized at typhoon-scale $3 \times 10^{-5}\,\mathrm{s}^{-1}$. External: IBTrACS v04r01 best-track fixes (≥ 34 kt where wind is reported), a triangle labeled positive if a fix falls within 1.5° of its centroid during the window. The external reference shares nothing with ERA5's assimilation of winds into our statistic — it is the community's independent record of where cyclones actually were. A synthetic Rankine-vortex test (`tests/test_geo.py`) verifies the whole bridge end-to-end: localization at the core, correct cyclonic sign after orientation correction, and rank agreement with the finite-difference vorticity.

S6 Reproducibility

All figures regenerate on CPU from fixed seeds via `experiments/run_all.py`; individually:

- Fig. 1 ← `run_fano.py` (pure theory, seconds);

- Fig. 2 $\leftarrow$ `run_partial_sampling.py` ($\approx$4 min);
- Fig. 3 $\leftarrow$ `run_plugin.py` ($\approx$3 min);
- the main paper's achievability grid and α-sweep $\leftarrow$ `run_second_order.py` ($\approx$15–25 min; K_4–K_8, random supports, full (N, ρ_2) grid, Wilson intervals, all cells in `results/second_order.json`);
- the main paper's vortex-localization study $\leftarrow$ `run_real_cyclone.py` ($\approx$3 min from the cached ERA5/IBTrACS files).

The test-suite (`pytest -q`, 52 tests) checks: the Fano bounds against the empirical recovery boundary of the phase-transition experiment (`run_phase_transition.py`, repo/supplement; a converse must lower bound it), the closed form (3) against Monte-Carlo, the envelope of Lemma S3.1, the adaptive-vs-known recovery gap, every claim of Section S4 (spark, exhaustive K_4 injectivity, separation 16, $C_G/\rho_2^2 \to 1$, Chernoff envelope, rank-deficient recovery), and the wind-to-edge-flow bridge (mesh full rank by Euler; a synthetic Rankine vortex localizes at the correct triangles with the correct sign and matches the finite-difference vorticity ranking).

References

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